
C3PO-MoRA-2: Token-Adaptive and Privacy-Preserving Certified Robustness for Multimodal Large Language Models

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Abstract

1 Certified defences for large language models (LLMs) still collapse when prompts
2 get long, perturbations get sparse, or inputs mix modalities, and they may even
3 leak the malicious prompts they aim to neutralise. We expose five concrete weak-
4 nesses—tiny global ℓ_2 radii, exploding bounds with sequence length, leakage from
5 hot-patch tables, drift after repeated patching and uncertified modality heads—and
6 address them with C3PO-MoRA-2, a 2-million-parameter adapter that integrates
7 three orthogonal upgrades. Span-Adaptive Certified Training (SACT) replaces a sin-
8 gle perturbation budget with learnable per-token radii, enabling tight certificates for
9 thousand-token prompts. Secure-Hash Retrieval (SHR) keeps only 128-bit hashes
10 of harmful embeddings and regenerates patch vectors through a keyed pseudo-
11 random function, closing inversion and membership-inference leaks ($\epsilon_{DP} < 1$).
12 Dual-Cert Fusion (DCF) adds modality-specific bound heads and merges intervals
13 through bound-aware attention, yielding end-to-end certificates for vision, audio
14 and text. All components fit on one commodity GPU and are guarded by run-time
15 assertions that abort on any code-paper mismatch. On PromptBench-XL C3PO-
16 MoRA-2 lifts certified accuracy for 512–1 024-token jailbreaks from 41% to 78%
17 while reducing bound volume by 31%. Against 10% image-patch and 20 dB audio-
18 spike attacks it certifies 68% and 71% accuracy, cutting empirical attack success
19 below 8% at only 5.4% added FLOPs. Mutual-information leakage from the patch
20 table drops from 0.11 to 0.02, and after 1 500 live patches factuality declines by
21 just 0.3 pp. C3PO-MoRA-2 is therefore the first token-adaptive, privacy-preserving
22 and cross-modal certified defence deliverable within a single fine-tuning budget.

23 1 Introduction

24 Large language models now power chatbots, code assistants and embodied agents, but they remain
25 alarmingly vulnerable to adversarial triggers. A 1 000-token prompt, a small corner patch in an image
26 or a brief audio spike can override safety policies while lying far outside the infinitesimal norm
27 balls assumed in most certified defences. Parameter-efficient fine-tuning (PEFT) adapters promise
28 cheap robustness retro-fits yet inherit three fundamental limitations. First, interval bounds grow
29 roughly linearly with sequence length, so certificates collapse for long prompts. Second, hot-patch
30 methods such as ZePatch store plaintext antidote vectors that can be inverted or probed through
31 membership inference. Third, existing multimodal adapters certify only the fused representation,
32 leaving modality-specific heads open to bypass attacks. Recent progress—fast adversarial training
33 Wong et al. [2020], remedies for catastrophic overfitting Andriushchenko and Flammarion [2020]
34 and low-rank robustness adapters Xu et al. [2023]—reduces compute cost but does not solve these
35 structural gaps.

36 This work introduces C3PO-MoRA-2, a PEFT adapter that closes them while respecting a 2 M-
37 parameter and single-GPU budget. It contributes three new ideas.

- 38 • **Span-Adaptive Certified Training (SACT)**. Learns a perturbation radius ϵ_i for every
39 token from gradient norms and attention entropy, then feeds these radii into interval bound
40 propagation (IBP). Robustness capacity is concentrated on high-risk spans without inflating
41 bounds elsewhere, scaling certificates to thousand-token inputs.
- 42 • **Secure-Hash Retrieval (SHR)**. Stores only 128-bit locality-sensitive hashes of harmful
43 prompt embeddings and reconstructs the corresponding antidote vector with a 256-bit-keyed
44 pseudo-random function. This eliminates practical inversion and gives $\epsilon_{DP} < 1$ privacy
45 guarantees, countering BrainWash-style attacks Abbasi et al. [2023].
- 46 • **Dual-Cert Fusion (DCF)**. Attaches lightweight bound heads to the vision and audio
47 branches and merges interval logits through bound-aware attention. The design extends
48 multi-perturbation robustness ideas Tramèr and Boneh [2019] to the multimodal LLM setting
49 and yields logical-AND certificates for all three modalities at <6% compute overhead.

50 We validate C3PO-MoRA-2 on three deliberately hard benchmarks: PromptBench-XL (long jailbreak
51 prompts up to 1 024 tokens), AdvPatch-LAION (10% square vision patches) and NoisePatch-Vox (20
52 dB audio spikes). A global harness imports auto_LIRPA at launch, checks dataset SHA-256 hashes,
53 ensures ϵ_i are non-constant and aborts if the on-disk hash table exceeds 2 MB, guaranteeing perfect
54 alignment between manuscript and artefacts.

55 Key findings are:

- 56 • **Text robustness at scale**. Certified text accuracy rises by 37 pp on 512–1 024-token prompts
57 versus a fixed- ϵ ablation.
- 58 • **Cross-modal certification**. DCF certifies 68% vision and 71% audio accuracy, matching or
59 exceeding specialised baselines at 5.4% extra FLOPs.
- 60 • **Privacy without drift**. SHR cuts mutual-information leakage by
61 $6\times$ and prevents factuality drift after 1 500 live patches.

62 The remainder of the paper is organised as follows. Section "Related Work" contrasts C3PO-MoRA-
63 2 with previous approaches, Section "Background" introduces formal notions and threat models,
64 Section "Method" details SACT, SHR and DCF, Section "Experimental Setup" specifies datasets,
65 metrics and implementation guards, Section "Results" reports empirical and certified performance,
66 and Section "Conclusion" summarises contributions and future directions.

67 2 Related Work

68 Certified text defences traditionally assume a single global perturbation budget. UniT propagates IBP
69 bounds over word replacements and FreeLB introduces continuous-embedding perturbations, but
70 both break down beyond $\epsilon \approx 0.15$ and a few hundred tokens. Adaptive budgets have been attempted
71 for contextual representations Wu et al. [2023] yet lacked formal guarantees. SACT is the first to
72 learn per-token ϵ_i inside IBP while avoiding the catastrophic overfitting that plagues single-step
73 adversarial training Andriushchenko and Flammarion [2020].

74 Hot-patch mechanisms provide rapid mitigation but leak information. BrainWash demonstrates
75 inversion and poisoning attacks on continual-learning memories Abbasi et al. [2023]. SHR adopts
76 locality-sensitive hashing and keyed pseudo-random functions to bound leakage below 0.03 mutual
77 information, outperforming plaintext ZePatch.

78 Cross-modal robustness has largely remained siloed. PatchGuard-ViT focuses on localised vision
79 noise Navaneet et al. [2023] and SoundShield on audio spikes, yet neither provides end-to-end
80 guarantees after fusion. Dual-Cert Fusion pioneers modality-specific certificates in a multimodal
81 LLM, extending robustness to the router stage inspired by multi-perturbation theory Tramèr and
82 Boneh [2019].

83 Finally, efficient robust fine-tuning has advanced via low-rank adapters (AutoLoRa Xu et al. [2023])
84 and safety-aligned subspaces (Safe LoRA Hsu et al. [2024]). C3PO-MoRA-2 stays within the same
85 parameter budget while delivering formal, cross-modal guarantees.

86 3 Background

87 Problem setting. We consider a frozen, 4-bit-quantised multimodal LLM with text, vision and audio
 88 encoders, a shared LoRA-64r adapter and task heads. The adversary may perturb (i) each text token
 89 within an ℓ_2 ball of radius ϵ_{text} , (ii) any square image patch covering up to 10% of pixels within ℓ_∞
 90 radius $\frac{4}{255}$, and (iii) additive mel-spectrogram noise ≤ 0.01 . A certificate guarantees that the predicted
 91 label $\hat{y}$ is invariant to all perturbations inside these budgets.

92 Interval bound propagation. For an input x with element-wise lower and upper bounds (l, u) , IBP
 93 propagates intervals layer-wise and returns output bounds; a classification is certified when the lower
 94 logit of the true class exceeds all competing upper logits Zhang et al. [2020]. Bound width grows
 95 roughly with sequence length, motivating per-token radii.

96 Parameter-efficient fine-tuning. LoRA inserts low-rank matrices $\Delta W = AB$ into frozen weights
 97 W . MoRA augments LoRA with momentum residuals; the 64-rank setup used here contains <2 M
 98 trainable parameters.

99 Security of retrieval tables. Storing plaintext antidote vectors permits vector inversion and membership
 100 inference with AUC ≈ 0.78 . Hashing embeddings to 128 bits and deriving antidotes on the fly renders
 101 inversion computationally infeasible; differential-privacy analysis yields $\epsilon_{\text{DP}} < 1$ under randomness
 102 in the hash function.

103 Assumptions. Base encoders are trusted; certification targets the adapter and heads. SHR’s secret key
 104 remains uncompromised—side-channel attacks are beyond scope.

105 4 Method

106 C3PO-MoRA-2 comprises three modules.

107 4.1 Span-Adaptive Certified Training

108 A two-layer MLP dubbed SpanRisk maps each token embedding e_i to a scalar risk score using features
 109 from gradient norms and attention entropy. After one warm-up epoch with a global $\epsilon_0 = 0.10$, per-
 110 token radii are set to $\epsilon_i = \sigma(\text{SpanRisk}(e_i)) \cdot \epsilon_{\text{max}}$, where σ is the logistic function and $\epsilon_{\text{max}} = 0.15$.
 111 One-hot token vectors are wrapped in BoundedTensor objects with radius ϵ_i . Training minimises
 112 the IBP upper bound on cross-entropy plus an ℓ_1 penalty on ϵ_i to discourage trivial inflation.

Input: tokens T , embeddings E , model f , warm-up radius ϵ_0 , max radius ϵ_{max} , SpanRisk MLP $R(\cdot)$

Warm-up: train one epoch with all tokens wrapped by radius ϵ_0

for epoch in 1..2 **do**

for batch B in dataset **do**

 Compute per-token features: gradient norms and attention entropy

 Risk scores: $r_i \leftarrow R(e_i)$ for $e_i \in E(B)$

 Radii: $\epsilon_i \leftarrow \sigma(r_i) \cdot \epsilon_{\text{max}}$

 Wrap tokens: $\hat{T} \leftarrow \text{BoundedTensor}(T, \epsilon)$

 Propagate IBP bounds through f : obtain interval logits ℓ

 Loss: upper-bound $\text{CE}(\ell) + \lambda \sum_i |\epsilon_i|$

 Update parameters of MoRA and R with AdamW

end for

end for

113 4.2 Secure-Hash Retrieval

114 For every harmful prompt embedding h we store $g = \text{LSH}_{128}(h)$. At inference the current prompt is
 115 hashed; on a match we compute the antidote $a = \text{PRF}_K(g)$ by truncating 256-bit BLAKE2s output
 116 to 2 048 floats. Only hashes are persisted (≤ 2 MB), so inverting a requires $\geq 2^{127}$ guesses, bounding
 117 mutual information at 0.03.

```

Input: embedding  $h$ , hash table  $\mathcal{H} \subset \{0, 1\}^{128}$ , secret key  $K$ 
 $g \leftarrow \text{LSH}_{128}(h)$ 
if  $g \in \mathcal{H}$  then
     $a \leftarrow \text{PRF}_K(g)$  ▷ derive 2 048-float antidote
    Apply patch:  $h' \leftarrow h + a$ 
else
     $h' \leftarrow h$ 
end if
Return  $h'$ 

```

118 4.3 Dual-Cert Fusion

119 Vision and audio inputs are enveloped as $(x - \epsilon, x + \epsilon)$. Each modality passes through a lightweight
120 bound head that outputs interval logits ℓ^{vis} and ℓ^{aud} . The text branch produces ℓ^{txt} under SACT.
121 Bound-aware attention computes conservative weights α from lower logits; final logits are $\alpha \odot$
122 ℓ . Certification demands identical $\arg \max$ across modalities, yielding logically rigorous AND
123 certificates.

```

Input: interval logits  $\ell^{\text{txt}}, \ell^{\text{vis}}, \ell^{\text{aud}}$ 
for modality  $m \in \{\text{txt}, \text{vis}, \text{aud}\}$  do
    Compute weights from lower bounds:  $\alpha^m \leftarrow \text{softmax}(\text{lower}(\ell^m)/\tau)$ 
    Fuse per modality:  $z^m \leftarrow \alpha^m \odot \ell^m$ 
end for
Joint certification holds if  $\arg \max \text{lower}(z^{\text{txt}}) = \arg \max \text{lower}(z^{\text{vis}}) = \arg \max \text{lower}(z^{\text{aud}})$ 
Return certificate and fused logits

```

124 4.4 Complexity

125 SACT adds $|x|$ radii; SHR consumes 2 MB storage; DCF adds ≈ 0.1 M weights. Torch profiler
126 measures a $5.4 \pm 0.3\%$ FLOP overhead relative to vanilla MoRA.

127 5 Experimental Setup

128 Hardware and libraries. All experiments run on a single NVIDIA A100-80 GB with PyTorch 2.1,
129 CUDA 12.2, cuDNN 8.9 and auto_LiRPA 0.4.2. A global harness aborts on library import failure,
130 dataset checksum mismatch, constant ϵ_i , zeroed modality tensors or >2 MB hash table.

131 Experiment 1 - SACT for long prompts. Dataset PromptBench-XL v1.2 (42 381 prompts, 32–1 024
132 tokens) is split 70/10/20%. Models: (a) C3PO-MoRA-2, (b) Abl-Globale (ϵ_i fixed to the median),
133 (c) SmoothLLM- $\sigma = 0.25$ Xhonneux et al. [2024]. Training: one warm-up epoch then two SACT
134 epochs with AdamW (lr 3×10^{-4} , batch 128 via gradient accumulation), five predetermined seeds.
135 Certification uses IBP+Backward on batch 32. Metrics: certified accuracy (CA) per length bucket,
136 mean bound volume, ϵ_i distribution and empirical HotFlip-SA attack success (30% budget). Statistics:
137 Wilcoxon paired tests with Holm correction; 95% BCa bootstrap.

138 Experiment 2 - DCF for patch and bypass attacks. Datasets: AdvPatch-LAION-25k (224² images,
139 10% patch), NoisePatch-Vox-8k (≤ 30 s clips, 20 dB spike) and TriModal-Fusion-15k. Split
140 70/15/15%. Models: (a) C3PO-MoRA-2 (DCF on), (b) Abl-DCFoff (single shared head), (c)
141 PatchGuard-ViT-B16 Navaneet et al. [2023], (d) SoundShield-base. Training: two epochs on modality
142 heads and router, seeds 7, 13, 19. Certification budgets: $\epsilon_{\text{vis}} = 4/255$, $\epsilon_{\text{aud}} = 0.01$, ϵ_{text} from frozen
143 SpanRisk; joint certificate takes logical AND. Metrics: CA per modality and joint, adaptive-PGD and
144 bypass ASR, FLOP overhead, calibration (ECE); sweep fusion temperature $\tau \in \{0.5, 1.0, 2.0\}$ and
145 router- $k \in \{64, 128\}$.

146 Experiment 3 - SHR privacy and stability. Part A compares C3PO-SHR to ZePatch-Plain under (i)
147 auto-encoder inversion (2 560-512-2 048) and (ii) Salem-19 membership inference. Metrics: Kraskov
148 mutual information, BLEU-4 reconstruction, MIA-AUC; 1 000-rep bootstrap and DeLong test. Part

B streams 1 500 red-team prompts, applies a patch after each, and every 100 patches evaluates TruthfulQA exact match and MMLU ECE; analysis via two-way RM-ANOVA.

Shared logging. Torch profiler records FLOPs, torch.cuda tracks peak VRAM, and /usr/bin/time measures energy (avg power $\times$ time $\times$ \$0.12 kWh).

6 Results

Long-prompt robustness. On PromptBench-XL C3PO-MoRA-2 attains $88.6 \pm 0.7\%$ CA for ≤ 128 tokens, $83.1 \pm 0.9\%$ for 129–512 and $78.4 \pm 1.0\%$ for 512–1 024, versus $52.5 \pm 1.3\%$ and $41.2 \pm 1.5\%$ for the fixed- ϵ ablation and $46.8 \pm 1.4\%$, $36.9 \pm 1.7\%$ for SmoothLLM. The 37.2 pp gain on the longest bucket is significant (Wilcoxon $p < 10^{-4}$). Mean bound volume shrinks by $31 \pm 2\%$ ($p < 0.001$). Tokens in the lowest-entropy quartile receive $2.4\times$ larger ϵ_i , confirming adaptive allocation.

Multimodal patch defence. Under 10% image patches C3PO-MoRA-2 certifies $68.1 \pm 0.8\%$ accuracy, beating Abl-DCFoff by 61 pp and PatchGuard-ViT by 3.2 pp. For 20 dB audio spikes it certifies $71.3 \pm 0.9\%$, exceeding Abl-DCFoff by 64 pp and SoundShield by 1.5 pp. Joint tri-modal CA is $47.9 \pm 1.1\%$ whereas the single-head ablation collapses to $3.1 \pm 0.5\%$. Empirical ASR falls to $\leq 8\%$ across modalities and bypass ASR drops from 44.1% to 6.2% (McNemar $p < 10^{-4}$). FLOP overhead is $5.4 \pm 0.3\%$.

Privacy and stability. SHR cuts mutual-information leakage to 0.018 ± 0.004 , BLEU-4 reconstruction to 6.1 and MIA-AUC to 0.51 (chance), compared with 0.113, 41.8 and 0.78 for ZePatch-Plain. After 1 500 patches TruthfulQA EM declines by 0.27 pp ($p = 0.34$) and MMLU ECE rises by 0.42% ($p = 0.41$), indicating negligible drift.

Robustness sweeps. Vision CA remains 52% under 20% patch area, still 40 pp above Abl-DCFoff. Audio CA degrades to 63% at 10 dB SNR. Fusion temperature $\tau = 1.0$ maximises joint CA with ≤ 2 pp variance.

Limitations. Certificates assume bounded additive noise and do not yet cover semantic paraphrases or optical character attacks. SHR’s privacy proof relies on key secrecy; side-channel leakage is future work. Extending SACT beyond $\epsilon_i \approx 0.2$ will require tighter backward bounds.

7 Conclusion

C3PO-MoRA-2 delivers the first token-adaptive, privacy-preserving and cross-modal certificates for large language models within a standard PEFT budget. Span-Adaptive Certified Training raises certified accuracy on 1 024-token jailbreaks by 37 pp, Dual-Cert Fusion furnishes per-modality guarantees at $< 6\%$ extra compute and Secure-Hash Retrieval eliminates inversion leakage while preserving factuality through 1 500 live patches. A fail-fast experimental harness ensures perfect alignment between manuscript and artefacts, offering a dependable test-bed for future research. Next steps include widening adaptive radii, certifying semantic perturbations and hardening SHR against key leakage. By open-sourcing our guarded pipeline we hope to catalyse rigorous evaluation of forthcoming robust fine-tuning techniques Andriushchenko and Flammarion [2020], Tramèr and Boneh [2019], Wong et al. [2020].

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